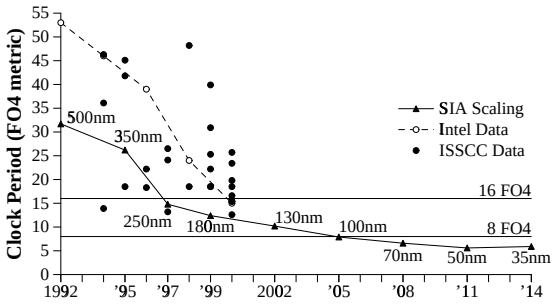


Figure 2: Clock scaling measured in FO4 inverter delays. The aggressive (8 FO4) and conservative (16 FO4) clocks are constant across technologies, but the SIA roadmap projects less than 6 FO4 delays at 50nm and below.

Gate (nm)	Chip Area (mm^2)	16FO4 Clk f_{16} (GHz)	8FO4 Clk f_8 (GHz)	SIA Clk f_{SIA} (GHz)
250	400	0.69	1.35	0.75
180	450	0.97	1.93	1.25
130	567	1.34	2.67	2.10
100	622	1.74	3.47	3.50
70	713	2.48	4.96	6.00
50	817	3.47	6.94	10.00
35	937	4.96	9.92	13.50

Table 2: Projected chip area and clock rate.

of technology scaling and deeper pipelining, with each improving approximately 15-20% per year. While the trend toward deeper pipelining will continue, reaching eight FO4 delays will be difficult, and attaining the SIA projected clock rate is highly unlikely.

In this paper, we examine microprocessor performance when scaling the clock conservatively at 16 FO4 delays (f_{16}), aggressively at eight FO4 delays (f_8), and to absolute limits with the SIA projections (f_{SIA}). In Table 2, we show the resulting clock rates across the spectrum of technologies we measure.

2.3 Wire Delay Impact on Microarchitecture

The widening gap between the relative speeds of gates and wires will have a substantial impact on microarchitectures. With increasing clock rates, the distance that a signal can travel in a single clock cycle decreases. When combined with the modest growth in chip area anticipated for high-performance microprocessors, the time (measured in clock periods) to send a signal across one dimension of the chip will increase dramatically. Our analysis below uses the clock scaling described above and the projected chip areas from the SIA Roadmap, as shown in Table 2.

Based on the wire delay model, we compute the chip area that is reachable in a single clock cycle. Our unit of chip area is the size of a six-transistor SRAM cell, which shrinks as feature size is reduced. To normalize for different feature sizes across the technologies, we measure SRAM cell size in λ, which is equal to one-half the gate length in each technology. We estimate the SRAM cell area to be $700\lambda^2$, which is the median cell area from several recently published SRAM papers [4, 23, 31]. Our area metric does not include overheads found in real SRAM arrays, such as the area required for decoders, power distribution, and sense-amplifiers. Additionally,

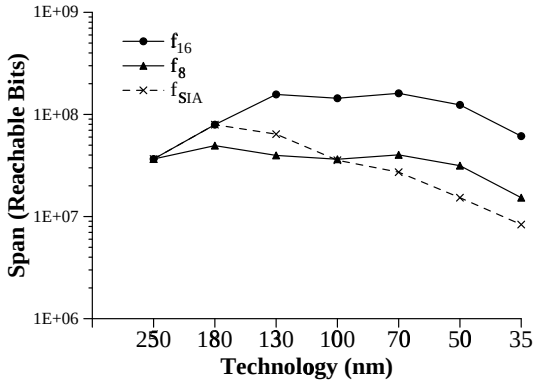


Figure 3: Reachable chip area in top-level metal, where area is measured in six-transistor SRAM cells.

it does not reflect the size of a single-cycle access memory array; the area metric includes all bits reachable within a one-cycle, one-way transmission delay from a fixed location on the chip, ignoring parasitic capacitance from the SRAM cells.

Figure 3 shows the absolute number of bits that can be reached in a single clock cycle, which we term *span*, using top-level wires for f_{16}, f_8, and f_{SIA} clock scaling. The wire width and spacing is set to the minimum specified in the SIA Roadmap for top-level metal at each technology. Using f_{16} clock scaling, the span first increases as the number of bits on a chip increases and the entire chip can still be reached in a single cycle. As the chip becomes communication bound at 130nm, multiple cycles are required to transmit a signal across its diameter. In this region, decreases in SRAM cell size are offset equally by lower wire transmission velocity, resulting in a constant span. Finally, the span begins to decrease at 50nm when the wire aspect ratio stops increasing and resistance becomes more significant. The results for f_8 are similar except that the plateau occurs at 180nm and the span is a factor of four lower than that of f_{16}. However, in f_{SIA} the span drops steadily after 180nm, because the clock rate is scaled superlinearly with decreasing gate length. These results demonstrate that clock scaling has a significant impact on architectures as it demands a trade-off between the size and partitioning of structures. Using high clock rates to meet performance targets limits the size of pipeline stages and microarchitectural structures, while tightly constraining their placement. If lower clock rates can be tolerated, then microarchitects can give less consideration to the communication delay to reach large and remote structures.

Figure 4 shows the fraction of the total chip area that can be reached in a single clock cycle. Using f_8 in a 35nm technology, less than 0.4% of the chip area can be reached in one cycle. Even with f_{16}, only 1.4% of the chip can be reached in one cycle. Similar results have been observed in prior work [16]. If microarchitectures do not change over time, this phenomenon would be unimportant, since the area required to implement them would decrease with feature size. However, microarchitectures have become more complex because architects acquired more transistors with each new fabrication technology, and used them to improve overall performance. In future technologies, substantial delay penalties must be paid to reach the state or logic in a remote region of the chip, so microarchitectures that rely on large structures and global communication